Ostéome ostéoïde : quelle est la meilleure option mini-invasive ? Comparaison Radiofréquence, Cryoablation, Micro-ondes et Laser

Titre original : Osteoid osteoma: which is the best mininvasive treatment option?

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1. Pertinence Clinique & Lien avec l'Agenda Interventionnel (Cochin)

Éclaire directement le choix entre Radiofréquence et Cryoablation (technique 'no-touch') pour les ostéomes ostéoïdes superficiels ou intra-articulaires comme celui de l'astragale pris en charge à Cochin.

2. Synthèse Clinique en Français

Résumé exécutif : Revue comparative des techniques percutanées d'ablation thermique de l'ostéome ostéoïde (RF, cryoablation, laser, micro-ondes). Alors que la RF reste la référence historique pour les lésions corticales classiques, la cryoablation offre un avantage majeur grâce à la visualisation directe du glaçon ('iceball') au scanner et à la possibilité d'une approche 'no-touch' extra-osseuse.

Analyse détaillée : Cette étude synthétise les performances cliniques et les profils de sécurité des modalités d'ablation percutanée de l'ostéome ostéoïde. L'efficacité globale dépasse 90 % quelle que soit la technique thermique, avec un taux de complications mineures inférieur à 3 %. La cryoablation se distingue par la possibilité de positionner la cryosonde au contact du cortex péri-lésionnel sans forer le nidus ('no-touch'), réduisant le traumatisme mécanique sur les petits os (talus/astragale, carpe) ou à proximité immédiate du cartilage et des nerfs.

3. Protocole Technique & Paramètres Opératoires

RF : canulation directe du nidus, 90 °C (4–6 min). Cryoablation : 1 sonde 17G en position juxta-corticale ('no-touch') ou intra-lésionnelle, cycle Congélation (8–10 min) – Décongélation passive/active (5 min) – Congélation (8–10 min) avec suivi TDM de l'isotherme 0 °C.

4. Points Clés à Retenir pour la Pratique

- Efficacité équivalente entre RF et Cryoablation (> 93 % de guérison clinique durable).
- La technique 'no-touch' en cryoablation évite le forage du nidus dans les os du tarse (astragale) et les lésions superficielles.
- Visualisation en temps réel de la zone d'ablation hypodense (iceball) sous scanner en cryoablation.
- Moins de douleurs post-procédurales immédiates après cryoablation grâce à l'effet analgésique intrinsèque du froid sur le périoste.

5. Original PubMed Abstract (Texte Intégral de l'Abstract)

Osteoid osteoma is the third most common benign bone tumor, with well-known clinical presentation and radiological features. Although surgical excision has been the only therapeutic option for a long time, to date it has been replaced by minimally invasive techniques, which proved satisfactory success rates and low complication occurrence. Therefore, the purpose of this literature review was to describe the main updates of these recent procedures in the field of interventional radiology, with particular attention paid to the results of the leading studies relating to the efficacy, complications, and recurrence rate. Nevertheless, this study aimed to analyze the peculiarities of each reported technique, with specific focus on the possible improvements and pitfalls. Results proved that all mininvasive procedures boast a high success rate with slight number of

complications and a low recurrence rate. Radiofrequency ablation is still considered the gold standard procedure for percutaneous treatment of osteoid osteoma, and it has the possibility to combine treatment with a biopsy. Interstitial laser ablation's advantages are the simplicity of use and a lower cost of the electrodes, while cryoablation allows real-time visualization of the ablated zone, increasing the treatment safety. Magnetic resonance-guided focused ultrasound surgery is the most innovative non-invasive procedure, with the unquestionable advantage to be radiation free.

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6. Full-Text Sections (Europe PMC Open Access JATS XML)

Introduction

Osteoid osteoma (OO) is the third most common benign bone tumor, representing approximately 3% of all primary bone tumors, and it mainly occurs between 5 and 25 years with a prevalence in the male population. The most affected site is the lower limb, particularly femur and tibia, and about 10% of osteoid osteomas arise in the vertebrae, primarily in the posterior elements [1–3].

Osteoid osteoma is radiologically and histologically characterized by a central hypervascular nidus of osteoblastic cells with a maximum diameter of 2 cm, surrounded by a sclerotic reaction, with low growth tendency [3].

Osteoid osteoma clinical presentation is extremely distinctive, represented by pain of variable intensity at the site of onset, that typically worsens during the night and improves with the administration of non-steroidal anti-inflammatory drugs [4].

Typical radiographic findings of osteoid osteoma include an intracortical radiolucent nidus associated with cortical thickening and reactive sclerosis, that at computed tomography (CT) examination manifests as a well-defined area of low attenuation with possibly central high attenuation focal spot representing mineralized osteoid [5].

Osteoid osteoma magnetic resonance imaging (MRI) findings include a hyperintense nidus with extensive adjacent bone marrow edema on fat-saturated sequences, characterized by contrast-enhancement after contrast medium administration [6]. Additionally, MRI better assesses OO relationship with nearby vascular and nerve structures, and it is a useful tool in the follow-up after percutaneous treatment [6, 7]. In fact, although CT can demonstrate the progressive dimensional nidus reduction and the cortical thickening resulting from ablation procedure, it is unable to evaluate post-procedural bone marrow edema and modifications in the adjacent soft tissues [8].

Technetium-99-labeled bone scintigraphy provides a metabolic assessment of the lesion, reflecting the radiotracer concentration at the site of increased osteoblastic activity, thus possibly supporting CT in the precise osteoid osteoma’s localization before treatment [9].

Although OO is a benign and self-limiting lesion, its spontaneous regression can take many years, causing patient’s protracted pain and forcing continuous and harmful use of anti-inflammatory drugs, with a significant deterioration in quality of life [2].

Surgical excision has been the only therapeutic option for a long time, albeit often associated with interventions, prolonged hospitalization, and significant risk of local recurrence in case of suboptimal resection. Over the years, less invasive treatment techniques have been developed with satisfactory success rates and improved post-treatment period [4].

Radiofrequency ablation (RFA)

Rosenthal was the first to successfully pave the way toward the use of radiofrequency for the treatment of OO. Since then, this technique has achieved excellent results in terms of efficacy and safety, so that percutaneous CT-guided radiofrequency ablation (CT-RFA) has become the gold standard therapeutic technique, replacing surgical treatment [10–12].

Furthermore, in comparison to surgery, RFA allows to reach difficult anatomical sites and provides shorter operation and hospitalization periods, with consequent costs reduction [12]. Specifically, Göksel et al. [13] carried out a comparative study on 24 patients with OO, 11 of which (45.83%) underwent surgical curettage, while 13 (54.17%) received RFA. Mean operation length of the surgery group was 69.5 min, with mean hospital stay of 1.3 days, while they were significantly shorter in the RFA group with mean operation length of 49.6 min and mean hospital stay of 0.3 days.

In technical terms, RFA is performed under CT guidance, necessary for lesion localization and electrode placement evaluation. Patients are usually under general, spinal, or propofol-induced anesthesia; local anesthesia alone is often associated with inadequate pain control [14].

The procedure consists in the insertion of an ablation electrode inside the osteoid osteoma nidus under CT guidance, followed by lesion heating, typically at 90 °C for a time of 5–6 min (Figs. 1, 2) [15].Fig. 1Osteoid osteoma of the right femoral neck. Axial (a) and coronal (b) MRI sequences weighted in STIR reveal the central nidus of the lesion (arrowhead) surrounded by extensive spongy bone edema and moderate joint effusion. c Axial CT scan shows the intracortical radiolucent nidus (black arrow) and the above cutaneous metal trace; d needle electrode positioning during radiofrequency thermal ablation procedure. Axial (e) and coronal (f) follow-up MRI sequences weighted in STIR after RFA demonstrate a sharp rim of post-procedural edema around the nidusFig. 2Osteoid osteoma of the left tibial diaphysis. Axial (a) and sagittal (b) CT scans show the intracortical radiolucent nidus with a central region of mineralization (arrowhead). c Axial CT scan performed during RFA procedure shows bone drill/biopsy kit over the cortical surface approaching the nidus

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The literature also describes ablation techniques involving different combinations of temperature and treatment duration: In particular, some authors have proposed a protocol in which instead of electrode direct overheating at 90 °C, there is an initial phase to raise the temperature to 90 °C, with a 2 min plateau at 60 °C, with a RFA-time longer than 15 min [15–19]. All of the authors achieved a success rate of over 90% and, in particular, Rimondi demonstrated that with this technique success improved from 79 to 98%, recurrences reduced from 21 to 2%, and complications from 5.9 to 0.2% ($p < 0.001$) compared to direct heating at 90 °C with an ablation lasting 4 min [18, 19].

Another RFA technique reported in the literature consisted in an ablative duration of 1–2 min with the electrode temperature at 90 °C, performed on 33 patients monitored for a mean follow-up of 92 months. The procedure was successful for 32 of 33 patients; RFA was repeated in one relapse occurred after 28 months and the patient was then free of symptoms [20].

Interstitial laser ablation (ILA)

Interstitial laser ablation (ILA) is a good choice among percutaneous treatments. An extensive revision of the literature (Table 3) revealed an overall success rate between 94 and 100% for the treatment of osteoid osteoma using ILA [62–70]. Some authors [62, 63] considered this technique easier to use than RFA because it does not require neutral electrodes and there is no current passing through the patient's body. A cost evaluation also showed that ILA is less expensive than RFA [62–65].Table 3ILA treatmentLead authorYearNumber of patientsNS/SSuccess (%)Follow-up in monthNumber of complicationsGangi2007114102/1299.158.51 mRoqueplan200910099/194241 M, 3 mWu20163636/094.4123

mEtienne20133594.3404 mWitt20002321/2100153 mZouari20081515/0100290Moser20086898830Rybak201 01313/010036.50Tsoumakidou20165757/0100120NS non-spinal osteoid osteoma, S spinal osteoid osteoma, M major complication requiring treatment, m minor complication.

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Gangi et al. [66] reported in a large study ILA performed from June 1994 to June 2004 in 114 patients with a mean age of 22.3 years, with an overall success rate of 99%. During the follow-up period of approximately 58.5 months, six cases of recurrence were described, two of which were related to imprecise needle positioning within the tumor nidus. All these patients were retreated with a second ILA, and only one case was unsuccessful.

Roqueplan [62] evaluated the efficiency and the complication rate of ILA, compared to percutaneous resection. This study reported a 94% of success rate at 24th month in a series of 100 patients, including 1 vertebral OO and 32 intra-articular lesions treated by ILA, and a 95% of success rate in 26 patients treated by percutaneous resection. In the group treated by ILA one procedure was considered technically unsuccessful, while three patients experienced recurrence. Totally four ILA were repeated, two of them failed again. In the second group, only one patient experienced no pain relief.

Similar to Gangi et al. [66], Roqueplan [62] registered that clinical failure was more frequent in patients younger than 16-year old. Low rate of complication was reported in the ILA group, about 4%: one major procedure-related complication due to the accidental injury of peroneal nerve, and three minor ones (hematoma, tendinitis, and infection). In the resection group, it is about 12%, with three minor complication: one case of meralgia paresthetica and cases of skin burns.

In 2016, Wu et al. [67] compared efficiency and complications between ILA and open surgery in a total of 72 patients divided into two groups, containing 36 patients each one. His experience has shown the superiority of percutaneous treatment over surgery, both in terms of clinical success rate (94 vs. 61%) and complication rate (8.3 vs. 27.8%). Despite the limitations of this study due to a limited number of patients and a short follow-up period (12 months), the superiority of percutaneous laser treatment over traditional surgery in terms of efficiency and safety was significant ($p < 0.05$).

Like Roqueplan [62], Etienne et al. [63] reported a clinical success rate about 94% in a series of 35 patients treated by ILA, including one vertebral localization. All procedures were considered technically correct, but two patients did not have pain relief and two patients had a recurrence (rate of 6%). These patients underwent a second procedure and only one had pain relief. Two of the remaining three underwent a third successful procedure. The complication rate in this study was about 5.4% including a skin burn, a patellar enthesopathy, a bone lacuna at the sacro-iliac joint, and a breakage of a needle during percutaneous procedure.

In the study by Witt et al. [68], the success rate was 100% in a group of 23 patients, including one case of recurrence efficiently treated with a second treatment. Only one vertebral lesion was included in the study. During the 15 months follow-up period, three minor complications were reported: Two patients had delayed healing of needle puncture, and one patient with distal phalanx osteoid osteoma lost the nail.

Microwave ablation (MWA)

Microwave ablation (MWA) is a relatively new technique, which has been growing rapidly in recent years and it provides some unique benefits compared to RFA, specifically less sensitivity to changes in tissue composition and bone impedance reaching higher temperatures within the tumor more rapidly and a more homogeneous volume of thermal ablation zone [71–75].

Moreover, MW provides an important advantage in the manage of those lesions not clearly visible or difficult to reach, since antennas do not need to be necessarily positioned exactly in the center of the nidus for an effective treatment [74].

However, for these same reasons, a recent review performed by Cazzato et al. claimed there are not enough evidences in the literature regarding the safety of MWA treatment on bone tumors, as bone tissue is susceptible to thermo-mediated complications such as secondary fractures [76].

Nowadays there are still not many studies available in the literature about this new technology concerning the treatment of osteoid osteoma (Table 4) [71–75, 77].Table 4MWA treatmentLead authorYearNumber of patientsNS/SSuccess (%)Follow-up in monthNumber of complicationsKostrzewa20141010/010060Basile2014 77/01005–130Rinzler20182424/010014 mPrud'homme20171313/092.313 mReis20201515/092.533.81 M, 2 mNS non-spinal osteoid osteoma, S Spinal osteoid osteoma, M major complication requiring treatment, m minor complication.

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The first two pilot studies published in 2014 are those of Kostrzewa [72] and Basile [73]; both reported 100% technical and clinical success rate over a follow-up period of 5 to 13 months and no complications were observed. However, both studies present the limit of restricted populations, respectively, 10 and 7 patients.

Rinzler et al. [77] published the largest study available in the literature, reporting the results of percutaneous MWA treatment in 24 patients with a 100% of clinical success, and 17% of complication rate, despite the limit of a short follow-up period, only 1-month.

Among the manuscripts available in the literature, only two reported a success rate less than 100%, respectively, the one of Prud'homme et al. with a success rate of 92.3% [74] and Reis et al. with 92% [75]. They studied a population of 13 and 15 patients, respectively, reporting in both cases only one clinical failure. Reis et al. [76] particularly compared the technical success, complication rates, and long-term efficacy or RFA and MWA therapies, considering two groups of 15 patients each. No statistical differences were noted between the group treated by RFA and the one treated by MWA, suggesting a comparable long-term recurrence rate ($p = 1,0$) and complication rates ($p = 0,60$) following RFA and MWA.

Cryoablation

Skjeldal et al. [78] were the first to apply cryoablation to an osteoid osteoma in correspondence with the ischial bone, demonstrating an optimal clinical success 1-year after treatment. In 2010, Liu et al. reported two cases of OOs treated by CT-guided cryoablation [79]. Through the years, this procedure has shown to offer advantages in those OOs near nerves, indeed motor nerve regeneration has been demonstrated following cryoablation reversible injury [15, 80]. Another advantage of cryoablation is the ability to identify the ablated zone as a low-density area which corresponds to the ice ball. Real-time monitoring of the ice ball allows to actively control the procedure and to minimize collateral damage to surrounding structures [15, 79, 81, 82].

Recent studies have demonstrated a further beneficial effect of cryoablation promoting an abscopal effect, an immune-specific reaction to tumor cell surrounding the ablation zone [15, 81].

Several trials reported high clinical success rate for percutaneous cryoablation of osteoid osteoma, ranging from 90.5 to 100% [82, 83].

Wu et al. considered six patients with a mean age of 12.6 years, showing a success rate about 100% without any case of recurrence neither complications. Coupal [83] and Santiago [84] considered a larger population of 10 (mean age: 27.9 years) and 21 patients (mean age: 29.9 years), respectively; Coupal et al. [83] reported a 100% of clinical success and no recurrences or complications. Santiago et al. [84] in 21 months of follow-up reported one recurrence and a final clinical success rate about 100% with a second successful cryoablation treatment. Three minor complications (14.3%) were encountered: one mild skin burn and two cases of soft tissue swelling and mechanical pain.

Whitmore [85] reported a 90.5% success rate in a series of 29 child patients (mean age: 11.3 years) reporting two clinical failures. He described two recurrences, both associated with juxtacortical positioning of the cryoprobe, one of them successfully treated by secondary RFA. Minor complication occurred in six patients

(21%): three cases of mild dermal blistering, two cases of weakness and pain, and one of transient numbness. All these cases were related to suboptimal positioning of the probe [85].

Only two studies [84, 85] evaluated axial skeleton, even if the study sample was small, one patient and five patients, respectively, with no difference in technical success and response to treatment compared to other lesions.

Percutaneous cryoablation can be considered a clinically effective and safe therapy, considering the low reported complication rate (range from 14.3 to 21%) (Table 5) [82–85] and the ability to provide a direct visualization of the ablation area under CT control (ice ball sign). Table 5Cryoablation treatmentLead authorYearNumber of patientsNS/SSuccess (%)Follow-up in monthNumber of complicationsWu201166/01002 8.70Coupal20141010/0100240Santiago20182116/5100213 mWhitmore20162928/190.5126 mNS non-spinal osteoid osteoma, S Spinal osteoid osteoma, M major complication requiring treatment, m minor complication.

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Magnetic resonance-guided focused ultrasound (MRg-FUS)

One of the most innovative techniques currently in use is magnetic resonance-guided focused ultrasound (MRg-FUS), which uses high-power ultrasonic energy to thermally coagulate body tissue [86]. It could be considered an attractive alternative to RFA, currently the gold standard.

MRg-FUS is a completely non-invasive procedure, based on the conversion of mechanical energy into heat, focused transcutaneously on a small tissue volume, without incisions, thus permitting the periosteal neurolysis and an immediate pain relief [86, 87] (Fig. 3).Fig. 3Osteoid osteoma of the left proximal tibial meta-diaphysis. a Technetium-99 m bone scintigraphy shows the typical double density sign, characterized by a central focus of intense radiotracer uptake (the nidus) within a region of generalized increase activity. b SPECT/CT multiplanar images demonstrate increased radiotracer uptake at the site of the nidus. c Axial MRI T2-weighted image with fat saturation during MRg-FUS shows treatment planning and osteoid osteoma location (arrowhead) relative to the transducer

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Two important advantages are the possibility of high-resolution imaging and the ability to estimate the level of heat generated in the tissue. The latter provides an indirect assessment of the effectiveness of nidus ablation and allows surrounding anatomical structures to be safeguarded by preventing burns [88].

Considering the high prevalence of osteoid osteomas in children and young adults, another relevant aspect concerns non-exposure to radiations and, since there is no mechanical penetration, even the rare risk of fracture or infection [87, 88].

However, accessibility to this new treatment is limited by certain factors, such as lesions not accessible to the MRgFUS system (e.g., spine OO), or general contraindications to perform MRI or administrate contrast medium. Furthermore, this treatment does not allow the biopsy of the lesion. Osteoid osteoma does not always need biopsy if it has a typical appearance. Another technique should be evaluated if an anatomical pathology evaluation is required (not-typical OO appearance) [87].

In literature, the clinical success rate of these new techniques is in a range between 87 and 100% (Table 6). In 2013, Napoli et al. [88] described six patients with osteoid osteoma. During 6 months of follow-up, a 100% of clinical success was reported, without any major or minor complications.Table 6MRg-FUS treatmentLead authorYearNumber of patientsNS/SSuccess (%)Follow-up in monthNumber of complicationsNapoli201366/010 060Napoli20174545/087360Geiger20142929/090120Arrigoni20193333/0100240Masciocchi20151515/093.324

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Conclusions

In conclusion, RFA is still considered the gold standard procedure for the percutaneous treatment of osteoid osteoma. Although RFA has the possibility to combine treatment with biopsy [15, 83], being a good alternative in cases of non-typical imaging appearance, preliminary biopsy execution is still a matter of debate among the authors, due to negative or inconclusive results. In fact, the usefulness of biopsy in the diagnosis of osteoid osteoma is still unclear, considering that clinical presentation and radiological findings are highly suggestive alone, so that histological examination is seen as unnecessary before therapeutic treatment [89].

Moreover, even if the advent of minimally invasive percutaneous CT-guided procedures made biopsy easier to access prior to intervention, the percentage of diagnostic biopsies reported in the literature remains relatively low (diagnostic yield range of 36–73%) [93–95, 97].

Similar to RFA, ILA and MWA can also be considered as viable percutaneous treatment alternatives with good results. ILA versus RFA is less expensive [52, 57, 58, 62] and presents minor technical concern [52, 57, 58].

MWA has shown promising results and some advantages over RFA such as a more homogeneous thermal ablation volume and a shorter treatment time [72–75]. Moreover, MW antennas do not need to be positioned exactly in the center of the nidus, helping in those lesions not clearly visible or difficult to reach [74].

Among percutaneous treatments, the only one that allows real-time visualization of the ablated zone (ice ball sign) is cryoablation, increasing the safety of treatment [72, 78, 81]. Furthermore, in case of unintended nerve injury, it is associated with lower risk of permanent nerve damage [80].

MRg-FUS is one of the most innovative treatments, the only non-percutaneous one considered in our review. The unquestionable advantage is the lack of radiation, an important factor given the prevalence of OO population [87, 89]. Another benefit is the ability to monitor the temperature in the treated area in real time, avoiding injury to surrounding tissues [88].

However, the main limitation of MRg-FUS is inaccessible locations (such as the spine). Moreover, the significant periosteal reaction that may occur near the osteoid osteoma nidus could be a limit to the US beam penetration, thus making superficial lesion more suitable to treatment [87, 88].

MRg-FUS does not allow the biopsy of the lesion, so in case of non-typical osteoid osteoma appearance, it is not the first-line treatment [87].